Northern University of Business and Technology, Khulna

Department of Computer Science and Engineering

CSE 4104 – Software Development III

AI-Based Software Project Development

Week 02 Assignment

Project Proposal

Project Title

KHABAR BONDHU (KindMeal)

AI-Powered Food Rescue and Need-Based Giving Platform

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1. Project Title

Khabar Bondhu (KindMeal): An AI-Powered Food Rescue and Need-Based Giving Platform

Khabar Bondhu (“Friend of Food”) is a web platform that rescues surplus food. It makes smaller institutions visible, to donors. Helps busy donors give with full transparency. The platform uses AI to check expiry dates match donors with institutions and provide Bangla-language support.

2. Problem Statement

1. What problem currently exists

Food that is still okay to eat gets thrown away every day while other people don't have enough to eat. This happens in two ways. First packaged food that is near the date when it should be thrown away stays on the shelves in bakeries and stores. Second after weddings, parties and other events there is often food left over. At the time many small madrasas and orphanages are not known to people who might want to help. These places get little support on a regular basis. Meanwhile kind people who want to give help often don't have the time to find an organization arrange the food deliver it and make sure it gets to the right place.

The amount of food that is wasted is well known. The UNEP Food Waste Index Report 2024 says that 19 percent of the food available to people around the world was wasted in 2022. That equals about 1.05 billion tonnes. Households were responsible for 60 percent of that waste food services for 28 percent and retail for 12 percent. For Bangladesh the same report estimates that households waste 14.1 million tonnes of food every year. That is 82 kg for each person. This Bangladesh number is an estimate based on a number of studies. It only includes food waste from homes not from stores or events. We use this number to show how big the

problem is. The amount of food that can be saved in our area is something we will measure during our first phase.

The fact that small institutions are not well known is what our team has noticed at first. We don't have survey data to support this yet. We will check this by doing a survey of institutions, in the city where we will start our project.

2. Why the problem is important

Food waste and hunger are happening at the time. Food waste and hunger are issues that need attention. The Global Hunger Index 2024 as reported by the Bangladeshi press placed Bangladesh in the hunger category. Bangladesh received a score of 20.4. Was ranked 75 out of 107 countries. Schools and kindergartens depend on help. When donors cannot find them or cannot check if they are receiving the help support goes to the well-known places and many other places are left out. Donors also feel unsure when they cannot be sure that their contribution was used properly. Trust and transparency are just as important, as the food itself.

3. Why existing solutions are insufficient

I started Week 01 research by looking at platforms such, as Too Good To Go, OLIO, MealConnect, Goodr and World Central Kitchen and I also examined Bangladeshi initiatives including Prochesta Foundation’s Project Food Banking, Ekjon Bangladesh and BhaatBank. International platforms are designed to give consumer discounts enable neighbour-to-neighbour sharing or work within existing food-bank networks. International platforms do not verify unregistered institutions and they do not manage donations on a donor’s behalf. Bangladeshi initiatives are valuable. Bangladeshi initiatives mostly depend on phone calls and volunteers. I found no evidence that Bangladeshi initiatives use verification, tracking or AI support. In my research I could not find a platform that brings together

packaged-food rescue, event surplus, verification of lesser-known institutions, donor-managed giving, transparent reports and Bangla-language AI support.

4. Why AI is relevant

AI helps solve three problems that a basic listing website can't handle. First checking product packages takes too long and often leads to mistakes. OCR can quickly read expiry dates from photos. Highlight items that are expired or close to expiring. Second matching donations to institutions is hard when there are many donations and too many institutions with different needs. Smart matching can rank institutions based on food type, amount, location, urgency and their recorded needs. Third many users at small institutions feel more comfortable using Bangla than English. A Bangla chatbot can guide them through the platform. AI supports and flags issues, but final decisions, about food safety always stay with a trained reviewer.

3. Objectives

The primary goals of the system are to:

- Create a web platform that connects food donors, verified institutions and donors who want to give but cannot manage the process themselves so that food donors can share their food easily.
- Reduce food waste by rescuing safe packaged food and event surplus and delivering it to institutions in need.
- Want to make known institutions visible, through verified profiles and a Need Map.
- Detect expiry dates using OCR and add safety flags that require review.
- Identify the organizations that match the donations and the budgets of the donors by using a feature that helps with the matching process.
- Provide a Bangla AI chatbot so that the non-technical users can use the platform in their own language.

- Provide transparency through records of collection, delivery and spending, with status updates to donors.
- Deliver a complete, deployed system with frontend, backend, database, authentication, AI integration, testing and documentation within the semester.

4. Proposed Features

Module / Area	Feature	Description
Core Platform	Authentication and roles	Registration and login for Donor, Food Partner, Institution, Volunteer and Admin, with password hashing and token-based sessions.
Food Rescue	Food listing and pickup scheduling	Partners (bakeries, shops, event organisers) list available food with type, quantity, expiry and pickup time; volunteers accept pickups.
Food Rescue	Safety review	Items flagged as expired, near-expiry or doubtful are held for review by an admin or trained reviewer before they are accepted.
Institution Connect	Institution profiles and verification	Profiles with needs and contact details, and three verification levels: Identified, Verified, Partner.
Managed Giving	Donation requests and tracking	A donor chooses an amount or purpose; the team fulfils it and the donor sees status, expense records and a completion report.
Need Map	Map of verified needs	OpenStreetMap view of which area has which verified need; only the general area is shown to protect privacy, especially children's.
Core Platform	Admin dashboard	Verify institutions, review flagged food, manage listings, and view activity records.

Module / Area	Feature	Description
Core Platform	Notification system	In-app or email updates when a listing is accepted, picked up, delivered, or a donation is completed.
AI Feature 1	OCR expiry-date detection	A photo of the package is processed to extract the expiry or best-before date and produce a safety flag.
AI Feature 2	Smart donor-to-institution matching	Ranks suitable verified institutions for a donation based on food type, quantity, location, urgency and need history.
AI Feature 3	Bangla AI chatbot	Answers questions in Bangla about donating, current needs and accepted food types, using platform data.

5. Technology Stack

Layer	Technology	Purpose
Frontend	React.js	Dashboards, forms, Need Map and chatbot interface
Main Backend	Node.js + Express.js	REST APIs, authentication, business logic
Database	PostgreSQL	Users, institutions, listings, donations, pickups and records
AI Backend	Python + FastAPI	Separate service for OCR, matching and chatbot calls
AI Integration	Gemini API or OpenAI API	Bangla chatbot and matching support
OCR	Python OCR library or OCR API	Expiry-date extraction from package images
Maps	OpenStreetMap	Need Map

Layer	Technology	Purpose
Version Control	Git + GitHub	Branches, collaboration and commit history
Deployment	Vercel or Netlify (frontend); Render or Railway (backend and AI service); hosted PostgreSQL	Live online deployment

API keys and database credentials will be kept in environment (.env) files and never uploaded to GitHub. The stack is initial and may be refined as development progresses.

6. Expected Outcome

1. What users will gain

- **Donors:** a trusted way to find verified institutions, give in the amount or form they choose, and receive proof that it was delivered.
- **Institutions:** visibility to donors and a more regular flow of food and support.
- **Food partners (bakeries, shops, organisers):** a simple way to pass on safe surplus instead of discarding it, with a record of what they contributed.
- **Volunteers and admins:** clear pickup tasks, schedules and a single dashboard for verification and review.

2. How the system improves the existing workflow

Step	Current practice	With Khabar Bondhu
Finding institutions	Word of mouth; small institutions stay unknown	Searchable verified profiles and a Need Map
Collecting food	Phone calls and informal coordination	Listings, scheduling and volunteer pickup in one place
Checking expiry	Manual look at each package	OCR reads the date; risky items are flagged for review

Step	Current practice	With Khabar Bondhu
Choosing a recipient	Personal judgement or familiarity	AI-assisted ranking by need, location and urgency
Proving delivery	Little or no record	Status tracking, records and completion reports

3. Expected benefits of AI integration

- Faster and more consistent checking of packaged food, which supports safer rescue decisions.
- Fairer and quicker matching, which reduces the chance that the same institutions are favoured repeatedly.
- Easier access for Bangla-speaking and non-technical users, which supports adoption by small institutions.

7. Scope and Feasibility

The semester version will be a working web platform (MVP) that includes the modules mentioned and the three core AI features. It will not include food-rescue route optimisation need analysis from institution requests, fraud and anomaly detection demand prediction, a donor recommendation assistant or AI impact analysis. These features are listed as work. Real-world operations like a field pilot with partner shops and institutions are extensions and are not required for the course.

The main risk is food safety. Because of this the platform does not claim that the AI can certify food as safe. Instead it flags items that may be questionable for review. Any doubtful food is rejected. If the platform moves into real-world use the team will follow the guidance of the Bangladesh Safe Food Authority and the Safe Food Act 2013.

Children's privacy is protected by showing the general area of institutions on the Need Map. Other risks, such as AI API costs or quotas and the accuracy of Bangla responses and OCR, on packaging will be handled by testing multiple prompts and tools during the AI integration and testing weeks.

References

- United Nations Environment Programme (2024). Food Waste Index Report 2024. Published 27 March 2024. unep.org.
- The Daily Star. "Why Bangladesh cannot afford to waste its food." Report on UNEP Food Waste Index 2024 findings for Bangladesh.
- The Financial Express (Bangladesh). "High household food wastage in BD: fact or myth?" Discussion of the UNEP estimate for Bangladesh.
- The Observer (Bangladesh). Reports on the UNEP Food Waste Index 2024 and Bangladesh's Global Hunger Index 2024 position.
- Week 01 submission: review of similar systems (Too Good To Go, OLIO, MealConnect, Goodr, World Central Kitchen, Prochesta Foundation, Ekjon Bangladesh, BhaatBank).